

Requirements:

- 1- Define a method called 'getPoint'. This method should return integer and take one argument which is the grade as character. This method should return the points of the grade. See the previous table to determine the points of each grade. [1 Point]
- 2- Define a method called 'getAllPoints'. This method should return integer and take one argument which is the grades as array of character. This method should return the total points of the all grades stored in the array. You must call 'getPoint' method to get the points of each grade. [2 Points]
- 3- Define a method called 'program2'. This method should not return anything and not take any arguments. This method should ask the user to enter the number of grades as integer. Then, this method should ask the user to enter each grade, and you have to store the grades in an array of character. After that, you must call 'getAllPoints' method to get the total points, also you have to calculate the average point of the grades. Then display the results as shown in the output example [3 Points]

Hints:

- 1- The average of the points = the total points / the count of the grades .

Main Method: [Total: 5 Points] + [Bonus: 1 Point]

You should display a menu and ask the user to choose a choice from the menu. The menu should continue be displayed after the user choice is executed except if the user chooses 0. The menu should look like as the following:

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CPCS 202 Final Lab Exam - A Model  
Enter 1 to start program 1.  
Enter 2 to start program 2.  
Enter 3 to show information about me.  
Enter 0 to exit.  
-----
```

Enter your choice:

Requirements:

- 1- Define a method called 'getUserSelection'. This method should return integer and take no arguments. This method should display the menu and ask the user to choose a choice. Then, the method returns the user's choice. [2 Point]
- 2- Define the main method, and call 'getUserSelection' method to display the menu and get the user choice. Then check the user's choice to start the chosen program. [1 Point]
- 3- If the user chooses number 3, you should display your full name and ID. [1 Bonus Point]
- 4- If the user chooses a number that is not in the menu, you should display this message "Incorrect choice, please try again". [1 Point]
- 5- If the user chooses number 0, you should display this message "Good Luck" and break the loop or stop the program. [1 Point]

Introduction:

You are going to make 2 simple and small programs, and those 2 programs are going to be represented in the main method as a menu that a user can select which program is to run.

Program 1: [Total: 4 Points]

Write a program that can calculate the distance between two points (p1X,p1Y) and (p2X,p2Y). Use the following equation to calculate the distance:

$$d(p1X, p1Y, p2X, p2Y) = \sqrt{(p1X - p2X)^2 + (p1Y - p2Y)^2}$$

Requirements:

- 1- Define a method called 'calcDistance'. This method should return 'double' and take appropriate arguments as integer. This method should calculate the distance between two points using the previous equation. [2 Points]
- 2- Define a method called 'program1'. This method should not return anything and not take any arguments. This method should ask the user to enter p1X, p1Y, p2X, and p2Y as integer. Then, this method should call 'calcDistance' method to calculate the distance of the given points and display the result to the user as shown in the output example. [2 Points]

Hints:

- 1- To calculate the power of a number, you can use Math.pow(a, b). The method calculates a^b and returns the result as double.
- 2- To calculate the square root of a number, you can use Math.sqrt(a). The method calculates $\sqrt{a}$ and returns the result as double.
- 3- You may need to import java.util.Scanner to use Scanner to get input from the user.
- 4- You may need to use 'System.out.printf' method to format the output.

Program 2: [Total: 6 Points]

Write a program that can calculate the points of grades and the average of the points of the grades. The points of each grade as the following:

Grade	Points
A or a	5
B or b	4
C or c	3
D or d	2
F or f	1
Other else	0

First, the program should ask the user to enter the number of grades. Then, you will ask the user to enter the grade for each grade. After that, you will display the results: total points of the grades and the average point of the grades.

Output Example:

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CPCS 202 Final Lab Exam - A Model
Enter 1 to start program 1.
Enter 2 to start program 2.
Enter 3 to show information about me.
Enter 0 to exit.
```

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Enter your choice: 1 Enter
Enter x for Point 1: 2 Enter
Enter y for Point 1: -4 Enter
Enter x for Point 2: 9 Enter
Enter y for Point 2: 3 Enter
The distance between the points (2,-4) and (9,3) = 9.90
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CPCS 202 Final Lab Exam - A Model
Enter 1 to start program 1.
Enter 2 to start program 2.
Enter 3 to show information about me.
Enter 0 to exit.
```

```
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Enter your choice: 2 Enter
Enter the number of grades: 5 Enter
Enter grade #0: a Enter
Enter grade #1: B Enter
Enter grade #2: f Enter
Enter grade #3: D Enter
Enter grade #4: d Enter
The total points of the grades: 14
The average points of the grades: 2.80
```

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CPCS 202 Final Lab Exam - A Model
Enter 1 to start program 1.
Enter 2 to start program 2.
Enter 3 to show information about me.
Enter 0 to exit.
```

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-----
Enter your choice: 3 Enter
Student Name: Your name
Student ID: Your ID
```

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CPCS 202 Final Lab Exam - A Model
Enter 1 to start program 1.
Enter 2 to start program 2.
Enter 3 to show information about me.
Enter 0 to exit.
```

```
-----
Enter your choice: 5 Enter
Incorrect choice, please try again.
```

```
-----
CPCS 202 Final Lab Exam - A Model
Enter 1 to start program 1.
Enter 2 to start program 2.
Enter 3 to show information about me.
Enter 0 to exit.
```

```
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Enter your choice: 0 Enter
Good Luck :)
```

Grading - Grade Yourself :

Task + Points	(For Student) Put your grade by yourself	(For Grader) Put the student grade
Program 1 (4 Points)	/ 4	/ 4
Program 2 (6 Points)	/ 6	/ 6
Main Method (5 Points)	/ 5	/ 5
Bonus (1 Point)	/ 1	/ 1
Total (15 Points) + (1 Bonus Point)	/ 15	/ 15

Good Luck 😊